

Leo Chen

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EDUCATION

Santa Clara University

B.S. Electrical and Computer Engineering

Santa Clara, CA

Sept 2022 – Jun 2025

Activities: Institute of Electrical and Electronics Engineers (IEEE), Association of Computing Machinery (ACM)

Coursework: Operating Systems, Computer Architecture, Embedded C & Assembly, Algorithms, Artificial Intelligence

EXPERIENCE

Software Engineer

Jun 2025 – Jun 2026

ProSomnus Sleep Technologies

Pleasanton, CA

- Spearheaded full-stack mobile application development for the ProSomnus RPMO2 (FDA Class II oral oximeter), integrating the cross-platform client with cloud backend services and the ProSomnus Oxymetrx ecosystem.
- Architected and deployed high-performance RESTful APIs in Go (Golang) hosted on AWS, designing MySQL schemas to securely ingest, process, and query multi-night patient physiologic telemetry.
- Engineered cross-platform Bluetooth Low Energy (BLE) communication protocols and built custom Python diagnostic tools, ensuring reliable, low-latency telemetry transmission across iOS, Android, Linux, and Windows.
- Implemented secure Over-The-Air (OTA) firmware upgrade workflows and automated internal software tooling, streamlining device manufacturing testing and deployment.
- Authored comprehensive automated test suites across mobile and backend services, ensuring robust code coverage and adhering to FDA medical device software standards (IEC 62304).

Undergraduate Machine Learning Researcher

Jun 2024 – May 2025

Santa Clara University

Santa Clara, CA

- Developed modular vision transformer models using TensorFlow, Keras, and PyTorch on Linux systems.
- Investigated Chain-of-Thought prompting and Backbone Branching to enhance detection transformer capabilities.
- Optimized detection transformers for satellite imagery via architectural layer modifications and hyperparameter tuning.
- Benchmarked models via ablation studies to improve inference throughput and object detection accuracy (mAP).

SKILLS

Languages: Go (Golang), Python, C/C++, SQL (MySQL), Dart, JavaScript/TypeScript, Unix Shell, Verilog

Cloud & Backend: AWS, MySQL, Podman(Docker)

Frameworks & Libraries: Flutter, BLE Protocols (Bleak), PyTorch, TensorFlow, Keras, Angular, NumPy, Matplotlib

Interests: Linux, Vim & Neovim, Open Source, Photography

PROJECTS & LEADERSHIP

Executive Board Member | Santa Clara University IEEE

Mar 2024 – Dec 2024

- Collaborated with professors and industry professionals to provide 200+ students with career & academic insights
- Designed quarterly university-wide software & firmware workshops to spark interest in electrical engineering
- Boosted club participation through a weekly newsletter and custom web pages reaching 400+ subscribers

Mentor | Santa Clara University

Aug 2023 – Jul 2024

- Provided structured technical guidance to incoming students, focusing on software engineering best practices, project design, and debugging strategies.
- Led one-on-one mentoring sessions focused on academic planning and career growth

Fist Bump | 3rd Place Hack for Humanity 2024

Feb 2024

- Developed a web platform to streamline student event tracking with a dynamic calendar view built in Dart
- Architected a RESTful backend using Dart's Shelf framework with 5 API endpoints connected to a local datastore for real-time event synchronization

The Marina Project | 3rd Place ACM Winter Challenge

Dec 2023

- Engineered a full-stack mobile application using Flutter and Dart
- Integrated 12+ REST APIs to curate music based on geographic location
- Optimized app responsiveness with data caching for a smooth user experience, improved response time by 30%
- Implemented asynchronous state management and skeleton loading to streamline page transitions and minimize perceptible latency